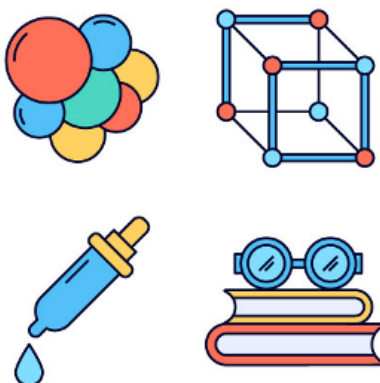




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational B • 2025



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. In the Fluid Mosaic Model of biological membranes, which of the following is not correct?
- A) The two leaflets of the membrane are symmetrical in both structure and function.
 - B) Phospholipids can move laterally and flip across to the other leaflet.
 - C) Glycoproteins are more likely to be embedded within the outer leaflet of the membrane.
 - D) Enzymes assist in the movement of phospholipids.
 - E) Phospholipids can congregate and distort the membrane structure.
- B02. Glands, such as the thyroid and adrenal glands, secrete chemical messengers that regulate bodily functions. Which bodily system is this?
- A) integumentary
 - B) nervous
 - C) cardiovascular
 - D) skeletal
 - E) endocrine
- B03. In terms of numbers, the *majority* of biochemical reactions in the cell use enzymes made of
- A) carbohydrates.
 - B) RNA.
 - C) DNA.
 - D) proteins.
 - E) lipids.
- B04. Blood, bone, and alveolar are _____ tissues.
- A) epithelial
 - B) integumentary
 - C) muscle
 - D) connective
 - E) nervous
- B05. Which of the following statements about DNA replication is incorrect?
- A) DNA replication occurs prior to cell division.
 - B) The replication of DNA produces RNA transcripts.
 - C) DNA polymerases replicate DNA.
 - D) Replication is bidirectional.
 - E) DNA replication occurs within the nucleus of eukaryotic cells and the nucleoid region of prokaryotic cells.
- B06. A biome that is dominated by heavy rainfall, hot temperatures (near the equator), and dense vegetation is
- A) temperate deciduous forest.
 - B) taiga.
 - C) tundra.
 - D) desert.
 - E) tropical rainforest.
- B07. Malaria-infected *Anopheles* inject _____ into humans during feeding.
- A) gametocytes
 - B) oocysts
 - C) ookinete
 - D) zygotes
 - E) sporozoites
- B08. Birds are most closely related to
- A) crocodilians.
 - B) fish.
 - C) mammals.
 - D) salamanders.
 - E) snails.

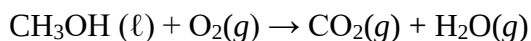
- B09. Consider the following genetic cross. What percent of the progeny will be heterozygous?
- Aa x AA
- A) 0%
 - B) 25%
 - C) 50%
 - D) 75%
 - E) 100%
- B10. Tapeworms, flukes, and roundworms are examples of
- A) endoparasites.
 - B) ectoparasites.
 - C) commensals.
 - D) mutualists.
 - E) biological vectors.
- B11. In oxygenic photosynthesis, _____ is the electron donor.
- A) oxygen
 - B) elemental sulfur
 - C) water
 - D) hydrogen sulfide
 - E) carbon dioxide
- B12. Which of the following mutations in DNA of a coding sequence would not result in an altered protein sequence?
- A) missense
 - B) silent
 - C) frameshift
 - D) nonsense
 - E) All of the above result in altered protein sequence.
- B13. The Centers for Disease Control and Prevention issued a Food Safety Alert in November 2024 for cucumbers and foods made with contaminated cucumbers, specifically for
- A) *Listeria*.
 - B) *Salmonella*.
 - C) *Escherichia coli* O27.
 - D) *Escherichia coli* O157:H7.
 - E) enterovirus.
- B14. Which of the following is a mechanism by which speciation can be prevented?
- A) natural selection
 - B) reproductive isolation
 - C) recombination
 - D) gene flow
 - E) mutation
- B15. What major event is occurring in S phase of the cell cycle?
- A) Condensation of the chromosomes
 - B) DNA replication
 - C) Cellular growth
 - D) Formation of the spindle apparatus
 - E) Division of the cytoplasm
- B16. Which of the following does not belong to Domain Eukarya?
- A) algae
 - B) mammals
 - C) hyperthermophiles
 - D) yeast
 - E) mushrooms

- B17. Consider the F₂ generation of Mendel's pea plants. If 16 progeny were produced in this generation from a dihybrid cross, statistically, how many of the 16 would be homozygous recessive for both traits?
- A) 1
 - B) 3
 - C) 6
 - D) 8
 - E) 16
- B18. Which major macromolecular group does ATP belong?
- A) proteins
 - B) carbohydrates
 - C) amino acids
 - D) lipids
 - E) nucleic acids
- B19. In a population at Hardy-Weinberg equilibrium, 450 out of 1000 individuals in the population have a recessive phenotype. Which Hardy-Weinberg variable is solved for first?
- A) p^2
 - B) p
 - C) q^2
 - D) q
 - E) $2pq$
- B20. The 30S ribosomal subunit of prokaryotes includes
- A) 46 proteins plus 28S, 5.8S, and 5S rRNA.
 - B) 33 proteins plus 18S rRNA.
 - C) 33 proteins plus 23S and 5S rRNA.
 - D) 19 proteins plus 16S rRNA.
 - E) the 50S ribosomal subunit.

C01. What is the molar mass of aluminum nitrate?

- A) 117.01 g/mol
- B) 213.01 g/mol
- C) 88.99 g/mol
- D) 184.99 g/mol
- E) 56.99 g/mol

C02. What is the sum of the coefficients when this reaction is balanced using the smallest whole number coefficients?



- A) 4 B) 6 C) 8 D) 9 E) 11

C03. Which of these statements is true when an electron in an oxygen atom jumps from $n = 2$ to $n = 5$?

- A) The oxygen atom is now an ion.
- B) The oxygen atom decays into a nitrogen atom.
- C) The oxygen atom has absorbed energy from its surroundings.
- D) The oxygen atom gives off a photon of light.
- E) This can't happen because oxygen is always O_2 and is never just one atom.

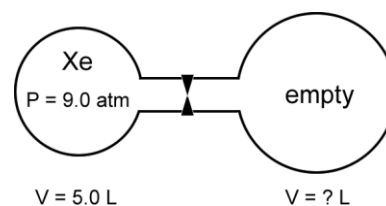
C04. Which of these is not a valid hybridized orbital type in valence bond theory?

- A) sp B) sp^2 C) sp^3 D) s^2p^3 E) sp^3d

C05. Only one of these is the correct name of an ionic compound. Which one is correct?

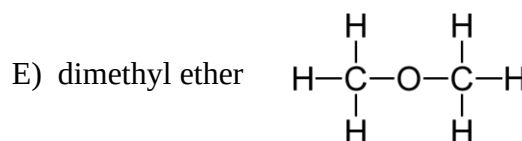
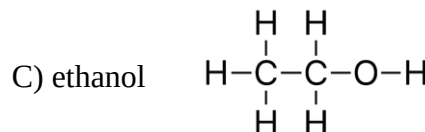
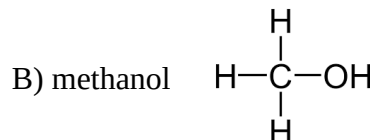
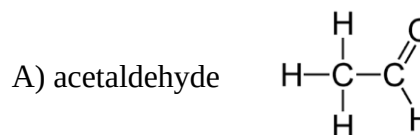
- A) Zinc(II) nitrate
- B) Copper(I) monochloride
- C) Dilithium oxide
- D) Silver(I) bromide
- E) Lead(II) iodate

C06. In the following two-bulb gas system, the bulb on the left has a volume of 5.0 L and contains 9.0 atm of xenon gas. The bulb on the right has an unknown volume and is empty. When the valve is opened allowing the xenon gas to fill both bulbs, the final pressure is 3.0 atm. What is the volume of the bulb on the right?



- A) 5.0 L B) 7.5 L C) 10.0 L
- D) 12.5 L E) 15.0 L

C07. Which of these liquids would you expect to have the highest boiling point?



- C08. A sample of solid hexamine, $\text{C}_6\text{H}_{12}\text{N}_4$, is burned, generating 3,000. J of heat. How many grams of hexamine were in the sample? The heat of combustion for hexamine is -4200.0 kJ/mol .
- A) 100.2 g B) 70.14 g C) 57.26 g
D) 151.63 g E) 95.83 g
- C09. A piece of an unknown metal with a mass of 50.14 g is heated to 88.0°C and dropped into 125 grams of water at a temperature of 25.0°C . If the final temperature of the water and the metal is 27.6°C , which of these could be the unknown metal?
- A) Chromium B) Tin
C) Titanium D) Tungsten
E) Zinc
- C10. When this reaction
- $$2\text{W}(\text{g}) + \text{X}(\text{g}) \rightarrow \text{Y}(\text{g}) + \text{Z}(\text{s})$$
- is allowed to reach equilibrium at 25°C , the amount of each substance at equilibrium is found to be
- W = 0.75 atm Y = 1.8 atm
X = 0.85 atm Z = 2.2 grams
- What is the value of the equilibrium constant for this reaction at 25°C ?
- A) 0.27 B) 3.8 C) 2.8 D) 0.12 E) 8.3
- C11. If you have 100 mL of 0.10 M $\text{Ba}(\text{OH})_2$ and you add 200 mL of water, what is the new pH of the solution?
- A) 1.48 B) 1.18 C) 13.00
D) 12.82 E) 12.52
- C12. What is the highest concentration of chloride ions you could have in a solution of 0.0010 M $\text{Pb}(\text{NO}_3)_2$ without forming a precipitate?
- A) 0.13 M B) 0.017 M
C) $1.7 \times 10^{-8} \text{ M}$ D) 0.25 M
E) 0.095 M
- C13. What is the oxidation number on the carbon atom in carbon monoxide?
- $:\text{C} \equiv \text{O}:$
- A) +2 B) -2 C) 0 D) +1 E) -1
- C14. How does a catalyst speed up a chemical reaction?
- A) It increases the kinetic energy of the reactants, resulting in more energetic collisions.
B) It creates a new reaction pathway with a lower activation energy.
C) It creates new attractions between the reactants, resulting in more interactions.
D) The presence of a catalyst tells the reactants you are serious about speeding up the reaction so they try harder.
E) It increases the value of the equilibrium constant for the reaction, resulting in the formation of more product.
- C15. How many molecules would there be in the gas phase if 325 mL of liquid methanol (CH_3OH) at room temperature is heated to 64.7°C and vaporized at 1 atm pressure? The density of liquid methanol at room temperature is 0.792 g/mL.
- A) 1.74×10^{24} molecules
B) 2.29×10^{24} molecules
C) 3.64×10^{24} molecules
D) 4.83×10^{24} molecules
E) 5.09×10^{24} molecules

C16. Consider the following four statements, which may be either true or false:

- I. Heat and temperature are the same thing.
- II. It is possible to add heat to something without changing its temperature.
- III. When you put ice in warm water, heat from the water is transferred to the ice and cold from the ice is transferred to the water.
- IV. Heat has negative mass and that is why heat rises.

How many of these statements are true?

- A) None of them
- B) One of them
- C) Two of them
- D) Three of them
- E) All of them

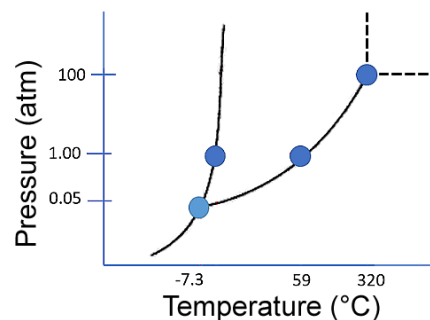
C17. What is the coefficient in front of O_2 when the chemical equation for the complete combustion of propane (C_3H_8) is balanced using the smallest integer coefficients?

- A) 1 B) 2 C) 3 D) 4 E) 5

C18. Your lab partner mistakenly pours 310 mL of 1.15 M NaCl solution into a jar containing 925 mL of 0.95 M KCl. What is the chloride ion concentration in the mixed solution?

- A) 0.95 M
- B) 1.00 M
- C) 1.05 M
- D) 1.10 M
- E) 1.15 M

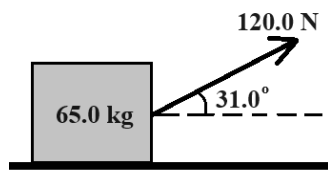
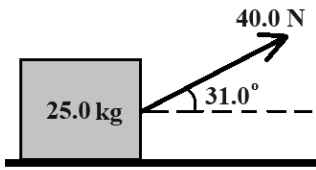
C19. What is the name of the point that appears at 320°C and 100 atm on this phase diagram?



- A) Absolute point
- B) Boiling point
- C) Critical point
- D) Demarcation point
- E) Extremity point

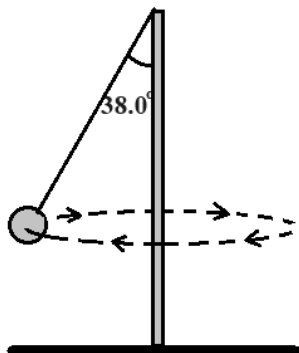
C20. If you had 165 grams of solid aluminum and you wanted to turn it all into AlF_3 by reacting it with HF gas, how many liters of HF gas would you need at STP to complete the reaction?

- A) 6.12 L
- B) 18.3 L
- C) 45.7 L
- D) 137 L
- E) 411 L

- P01. According to Orzel, the results from both Maxwell's equations and the Michaelson-Morley experiment forced physicists to reconsider some deeply held notions. The most fundamental notion that had to be abandoned was the idea of...
- an absolute frame
 - the aether
 - simultaneous length measurement
 - relative velocity
 - universal time
- P02. According to Orzel, a particular type of subatomic particle that is created by cosmic rays in the upper atmosphere has a very short lifetime and should decay before reaching the ground. However, due to the high speed of these particles, and the clock-slowness effects of relativity, these particles are able to reach the surface of the Earth where they are detected in substantial numbers. What are these particles?
- muons
 - electrons
 - pions
 - D-mesons
 - tauons
- P03. According to Orzel, measuring the length of an object moving at relativistic speeds can be reduced to a problem of...
- the length measurement tools
 - reference frame correlation
 - reference frame acceleration
 - the synchronization of clocks
 - multiple observers
- P04. Uranus and Neptune are called "ice giants" because of their large mantle regions composed of warm, dense ice. This ice makes up about 80% of the mass of these planets. What are the primary three compounds that form the ice in Uranus and Neptune?
- water, ammonia, methane
 - water, carbon dioxide, ammonia
 - water, hydrogen, methane
 - ammonia, hydrogen, methane
 - ammonia, hydrogen, carbon dioxide
- P05. An electric field lies entirely in the x-y plane. The electric field is oriented at an angle of 35.0° with respect to the positive x-axis, and the component of the field in the x-direction has a magnitude of 140.0 N/C . What is the magnitude of the total electric field?
- 140 N/C
 - 171 N/C
 - 200 N/C
 - 244 N/C
 - 345 N/C
- P06. A truck starting from rest accelerates at a constant rate of 2.25 m/s^2 until it reaches a speed of 30.0 m/s . The truck then travels at constant speed for 5.00 seconds. At that point, how far has the truck traveled in total?
- 150 m
 - 200 m
 - 350 m
 - 400 m
 - 550 m
- P07. You pull a 65.0 kg crate to the right with a 120.0 N force, as shown. Your force acts on the crate at an angle of 31.0° above the horizontal, and the floor on which the crate slides is frictionless. What is the acceleration of the crate?
- 0.542 m/s^2
 - 0.951 m/s^2
 - 1.11 m/s^2
 - 1.58 m/s^2
 - 1.85 m/s^2
- 
- P08. You decide the crate from problem P07 is too heavy, so you switch to pulling a 25.0 kg crate. You pull with a force of 40.0 N , at an angle (once again) of 31.0° . The floor is frictionless. If you pull the crate a horizontal distance of 45.0 m , how much work have you done?
- 927 J
 - 1082 J
 - 1543 J
 - 1800 J
 - 3857 J
- 

- P09. A 3.40kg ball is attached to the end of a 1.50m long rope that is connected to the top of a vertical pole. The ball is kicked so that it spins around the pole in a circle (as shown). While spinning, the rope makes an angle of 38.0° with respect to the pole. What is the angular velocity of the ball as it spins around the pole?

- A) 2.88 rad/s
B) 5.52 rad/s
C) 8.28 rad/s
D) 8.98 rad/s
E) 12.5 rad/s



- P10. You attach a mass to a string to form a simple pendulum. Your pendulum has a period of 0.750 seconds. You enter an elevator and while the elevator is accelerating upwards, the period of the pendulum changes to 0.650 seconds. What is the acceleration of the elevator?

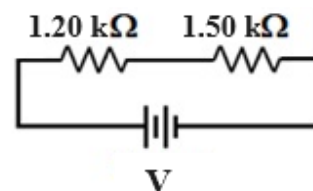
- A) 0.727 m/s^2
B) 1.51 m/s^2
C) 2.44 m/s^2
D) 3.25 m/s^2
E) 7.36 m/s^2

- P11. A 20.0g piece of aluminum that has a specific heat capacity of 900 J/kgK is heated until it glows. The aluminum is placed in a cup containing 100.0g of pure water with an initial temperature of 22.0°C . The specific heat capacity of water is 4186 J/kgK . After reaching thermal equilibrium, the water and aluminum have a final temperature of 55.0°C . What was the temperature of the aluminum when it was glowing?

- A) 602°C
B) 712°C
C) 822°C
D) 932°C
E) 1040°C

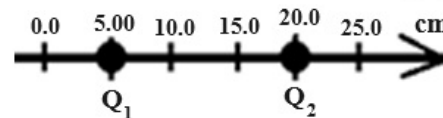
- P12. For the circuit shown below, the energy dissipated in the $1.20\text{k}\Omega$ resistor is 0.2016W . Determine the magnitude of the voltage source, V .

- A) 11.0 V
B) 17.6 V
C) 24.2 V
D) 30.2 V
E) 35.0 V



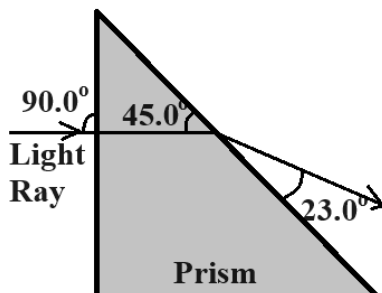
- P13. A $4.50\mu\text{C}$ charge (Q_1) is placed on the x-axis, at the 5.00cm mark, as shown below. Calculate the work needed to move a second charge, $Q_2 = 1.25\mu\text{C}$, from a long distance to the 20.0cm mark on the axis.

- A) 0.253 J
B) 0.337 J
C) 1.26 J
D) 2.25 J
E) 5.06 J



- P14. Aluminum is an example of a paramagnetic material. What does that tell us about the magnetic dipoles of the atoms in aluminum?
- A) The dipoles align with an external magnetic field but do not remain aligned when the external field is removed.
B) The dipoles anti-align with an external magnetic field but do not remain anti-aligned when the external field is removed.
C) The dipoles align with an external magnetic field and remain aligned when the external field is removed.
D) The dipoles anti-align with an external magnetic field and remain anti-aligned when the external field is removed.
E) The dipoles are unaffected by an external magnetic field.

- P15. A light beam intersects a prism as shown below. The beam enters the prism at 90.0° to the first surface. It passes through the prism and intersects the second surface at an angle of 45.0° . The beam refracts, leaving the prism at an angle of 23.0° with respect to the second surface. Based on this information, determine the index of refraction of the prism.



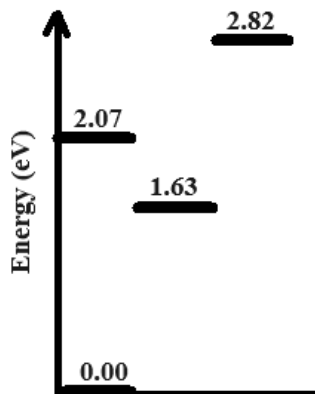
- A) 2.56
B) 1.81
C) 1.60
D) 1.41
E) 1.30

- P16. You use a converging lens as a magnifier to examine a small snail. The snail is located 6.00 cm in front of the lens, and its actual size is 1.22 mm. The image of the snail that you see using the magnifier is upright and 3.75 mm in size. What is the focal length of the lens?

- A) 1.95 cm
B) 3.07 cm
C) 4.53 cm
D) 6.00 cm
E) 8.89 cm

- P17. You have discovered a new element which has the energy levels illustrated in the diagram below. Which of the following wavelengths of light would not be produced by an atom of this element?

- A) 335 nm
B) 440 nm
C) 599 nm
D) 761 nm
E) 1042 nm

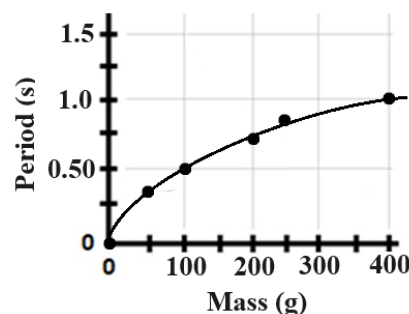


- P18. The half-life of Carbon-14 is 5730 years. You discover a bone fragment in which the Carbon-14 content has decreased to 12.5% of its original amount. Approximately how old is the bone fragment?

- A) 11500 years
B) 17000 years
C) 23000 years
D) 32500 years
E) 46000 years

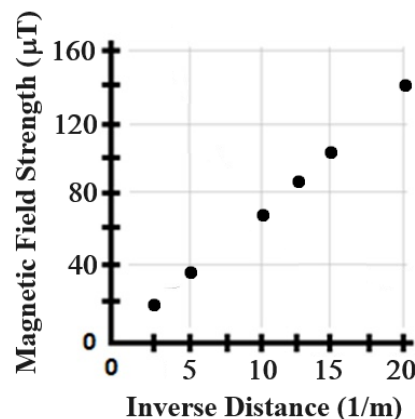
- P19. You attach different masses to a spring, and each time set the system into harmonic motion. For each mass, you measure the period of the harmonic motion. A plot of the period, T , versus the mass, m , is shown below. From these data, determine the spring constant of the spring.

- A) 40 N/m
B) 25 N/m
C) 16 N/m
D) 7.1 N/m
E) 2.5 N/m



- P20. A long straight wire carries an unknown current, I . You measure the strength of the magnetic field produced by the current at different distances from the wire. Then you plot the magnetic field strength, $|B|$, versus the inverse of the distance $1/r$. The plot is shown below. From these data, determine the magnitude of the current flowing in the long straight wire.

- A) 5.6 A
B) 18 A
C) 25 A
D) 35 A
E) 49 A



1A 1	Chemistry																8A 18
1 H 1.01	2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp}	= 0°C
T_{bp}	= 100°C
c_{ice}	= 2.09 J/g·K
c_{water}	= 4.184 J/g·K
c_{steam}	= 2.03 J/g·K
ΔH_{fus}	= 334 J/g
ΔH_{vap}	= 2260 J/g
K_f	= 1.86 °C/m
K_b	= 0.512 °C/m

Constants

R	= 0.08206 L·atm/mol·K
R	= 8.314 J/mol·K
R	= 62.36 L·torr/mol·K
e	= 1.602×10^{-19} C
N_A	= 6.022×10^{23} mol ⁻¹
k	= 1.38×10^{-23} J/K
h	= 6.626×10^{-34} J·s
c	= 3.00×10^8 m/s
R_H	= 2.178×10^{-18} J
m_e	= 9.11×10^{-31} kg
$\mathcal{F}$	= 96,485 C/mol e ⁻
1 amp	= 1 C/sec
1 mol e ⁻	= 96,485 C

Specific heats of some metals (J/g°C)

Chromium	0.449
Tin	0.227
Titanium	0.523
Tungsten	0.134
Zinc	0.387

 K_{sp} for PbCl₂

$$K_{\text{sp}} = 1.7 \times 10^{-5}$$

Physics Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles $\rightarrow$ meters	$1.00 \text{ mile} = 1609 \text{ meters}$
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3